nature physics

Article

https://doi.org/10.1038/s41567-025-03095-1

Twisting the Hubbard model into the momentum-mixing Hatsugai–Kohmoto model

Received: 4 September 2024

Accepted: 8 October 2025

Published online: 27 November 2025

Check for updates

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The Hubbard model is a standard theoretical tool for studying materials with strong electron–electron interactions, such as cuprate superconductors. Unfortunately, interaction-driven phenomena, such as a transition into the strongly correlated Mott insulator phase, are difficult to treat with established theoretical techniques. However, the exactly solvable Hatsugai–Kohmoto model displays similar Mott physics. Here we show how the Hatsugai–Kohmoto model can be deformed continuously into the Hubbard model. The trick is to systematically reintroduce all the momentum mixing that the original Hatsugai–Kohmoto model omits. This can be accomplished by grouping n momenta into a cell and hybridizing them, resulting in the momentum-mixing Hatsugai–Kohmoto model. We recover the Bethe ansatz ground-state energy of the one-dimensional Hubbard model to within 1% from only ten mixed momenta. Overall, the convergence scales as $1/n^2$ as opposed to the inverse linear behaviour of standard finite-cluster techniques. Our results for a square lattice reproduce all the known features from state-of-the-art simulations also with only a few mixed momenta. Consequently, we believe that the momentum-mixing Hatsugai–Kohmoto model offers an alternative tool for strongly correlated quantum matter.

The paradigmatic model for strong correlations in quantum matter is that of electrons hopping on a lattice between nearest neighbours but paying the energy cost U anytime they doubly occupy the same site. Naively, the energy eigenstates of this model would seem to be ordered with respect to the number of doubly occupied sites. However, this fails because the kinetic and potential terms do not commute. As a result, electrons can lower their energy by hopping to sites with single occupancy. It is for this reason that this simple model is unsolved, except in $d = 1$ (ref. 1) and $d = \infty$ (ref. 2), thereby rendering the $d = 2$ Mott problem a grand challenge as it is the gold standard for Mott physics in cuprates. The state-of-the-art methods remain quantum Monte Carlo (QMC)[3–7], density matrix renormalization group (DMRG)[3–5,8] and dynamical mean-field theory (DMFT)[2,9,10] and its cluster extensions such as cellular DMFT[11,12] and the dynamical cluster approximation[13–16], as well as diagrammatic extensions[7,17] such as dynamical vertex approximation[18,19] and dual fermion[20], which have all been useful in unveiling the properties of the Hubbard model including pseudogap[13,19–24], superconductivity[5,13] and transport[10,25,26] in the strange-metal regime.

A natural question arises: is there an alternative that allows a natural interpolation between the two solvable limits of $d = 1$ (ref. 1) and $d = \infty$ (ref. 2). It is this problem that we tackle here. The more tractable formulation of Mott physics is the Hatsugai–Kohmoto (HK) model[27–32], which we hereafter refer to as the band HK model. It has a repulsion

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between any two electrons with opposite spin that doubly occupy the same momentum state. Consequently, the doubly occupied band steadily moves upwards in energy as the repulsion increases, resulting in an insulating state in a half-filled band when the interaction strength exceeds the bandwidth. In this model, double occupancy is the organizing principle for ordering of the eigenstates. Clearly then, this is a gross simplification of the Hubbard model stemming from the locality in momentum space (long-range interactions in real space) and generating a macroscopic spin degeneracy in the singly occupied sector. In essence, the solvability of band HK rests on the commutativity of the kinetic and interaction terms. Can such non-commutativity be put back in systematically, thereby reducing the difference from the Hubbard model, without giving up on exact solvability completely?

We show here that this can be done by constructing the momentum-mixing Hatsugai–Kohmoto (MMHK) model. The procedure is to group the momenta into cells and then mix them. For example, grouping the **k** states into cells of two mixes all momenta differing by **q** = (0, 0) and (π, π). More generally, for an integer n we mix n momenta related by a fixed set of translation vectors $\{\mathbf{q}_1, ..., \mathbf{q}_n\}$, which we denote as the n-MMHK model. We show that just a few mixing momenta are needed to recover all known results in one dimension[1] and is in agreement with the standard results in two dimensions for the Mott transition[2,13], spin susceptibility[33], spectral function[34,35], double occupancy[33] (H. Xiu et al., unpublished data), dynamical spectral weight transfer (DSWT)[36,37] and heat capacity[38,39]. We show that in one dimension, the convergence to Hubbard physics in the thermodynamic (TD) limit scales as ~$1/n^2$, whereas in finite-cluster Hubbard calculations, the convergence as a function of system size is inversely linear. The nature of rapid convergence is explained by a scaling argument in the penultimate section and we show that it becomes even faster in higher dimensions because quantum fluctuations are suppressed. In essence, the MMHK model is an efficient simulator of Mott/Hubbard physics, enabling analytical insights and numerical augmentation. This is a consequence of the flow of MMHK to the Mott fixed point in which a discrete Z_2 symmetry is broken[29].

Towards Hubbard

At the outset, it is helpful to write the explicit form of the Hubbard and HK models. Although both models share the same kinetic energy, $\sum_{\mathbf{k},\sigma}\xi_{\mathbf{k}}n_{\mathbf{k}\sigma}$, their potential energy terms

$$\begin{aligned} H_{\text{int}}^{\text{HB}} &= U\sum_i n_{i\uparrow}n_{i\downarrow} \\ &= \frac{U}{N}\sum_{\mathbf{k},\mathbf{p},\mathbf{q}\in\text{BZ}} c^{\dagger}_{\mathbf{k}\uparrow}c_{\mathbf{k}-\mathbf{q}\uparrow}c^{\dagger}_{\mathbf{k}+\mathbf{p}\downarrow}c_{\mathbf{k}+\mathbf{p}+\mathbf{q}\downarrow} \end{aligned} \tag{1}$$

and

$$H_{\text{int}}^{\text{HK}} = U\sum_{\mathbf{k}\in\text{BZ}} n_{\mathbf{k}\uparrow}n_{\mathbf{k}\downarrow} \tag{2}$$

differ substantially. In Hubbard, antiparallel spin electrons with local on-site occupancy, $n_{i\sigma}$, repel one another, whereas in HK, doubly occupying the same momentum state turns on the interaction. The occupancy in each momentum state is $n_{\mathbf{k}\sigma}$ and $\xi_{\mathbf{k}}$ is the energy of each band state with momentum **k**. It is easy to see that because the HK model is diagonal in momentum space, when U exceeds the bandwidth set by $\xi_{\mathbf{k}}$, a gap opens at the chemical potential and the spectral weight will now be carried by two bands each with equal weight at half-filling. It is this bifurcation of the spectral weight that is central to Mott physics, as depicted in figure 1 of ref. 40. Figure 1a demonstrates that the density of states (DOS) for various U values across the Mott transition of the half-filled band HK model always exhibits a finite DOS at $\omega = 0$ for $U < W$. This central peak only vanishes in the insulating phase. In HK, there are four states per **k** point. None of the **k** points are hybridized.

To bridge the gap with Hubbard, we must introduce hybridization. The simplest way to bridge this gap is to put two **k** points into a unit cell, and then hybridize them. The difference between the hybridized momenta will be $B_2 \equiv \{(0, 0), (\pi, \pi)\}$. In this basis, the effective interaction can be written in the standard Hubbard form as

$$H_{\text{int},n=2}^{\text{MMHK}} = \frac{U}{2}\sum_{\mathbf{k}\in\text{BZ}}\sum_{\mathbf{p}\in\mathbf{B}_2}\sum_{\mathbf{q}\in\mathbf{B}_2} c^{\dagger}_{\mathbf{k}\uparrow}c_{\mathbf{k}-\mathbf{q}\uparrow}c^{\dagger}_{\mathbf{k}+\mathbf{p}\downarrow}c_{\mathbf{k}+\mathbf{p}+\mathbf{q}\downarrow}. \tag{3}$$

However, by bipartitioning the lattice and introducing the basis for each unit cell (sums and differences of the **k**-state fermionic operators), $c_{\mathbf{k}A\sigma} = \frac{1}{\sqrt{2}}\left(c_{\mathbf{k}\sigma} + c_{\mathbf{k}+(\pi,\pi)\sigma}\right)$ and $c_{\mathbf{k}B\sigma} = \frac{1}{\sqrt{2}}\left(c_{\mathbf{k}\sigma} - c_{\mathbf{k}+(\pi,\pi)\sigma}\right)$, we find that this potential can be written in a compact form:

$$H_{\text{int},n=2}^{\text{MMHK}} = U\sum_{\mathbf{k}\in\text{rBZ}_2}\left(n_{\mathbf{k}A\uparrow}n_{\mathbf{k}A\downarrow} + n_{\mathbf{k}B\uparrow}n_{\mathbf{k}B\downarrow}\right), \tag{4}$$

reminiscent of the HK interaction in equation (2) but with **k** and **k** + (π, π) mixed ($n = 2$). In doing so, we must restrict the momentum summation to the reduced Brillouin zone (rBZ$_2$), which is exactly half the original Brillouin zone (BZ). Then, for $n = 2$, we hybridize the **k** points that differ by (π, π), thereby lifting their degeneracy. This procedure, which can be repeated with any size of unit cell, continuously deforms HK into the Hubbard model. The question we ask is can this procedure be automated and how large does the unit cell have to be to reproduce non-trivial Hubbard physics?

MMHK

We now outline how the **k**-point hybridization picture can be automated. As pointed out previously[41], band HK can be generalized to include multiple orbitals per **k** state as necessary for the adaptation of this model to topological models, all of which have at least two atoms per unit cell[42–44]. Such a change produces qualitatively different physics as the hybridization between the orbitals lifts[41,44,45] the TD degeneracy of the band HK model and preserves only the degeneracy constrained by symmetry[46]. The procedure we outline here is equivalent to this scheme only if the number of degrees of freedom is fixed as orbitals are added. That is, the BZ with n orbitals per site is reduced by a factor of $1/n$. Since there is no constraint of this kind inherent in the orbital model[41], we refer to the scheme adopted here as the MMHK model. As shown later, the convergence of MMHK to the Hubbard physics in the TD limit scales as fast as ~$1/n^2$.

Since we are only interested in Hubbard physics for now, we consider an MMHK model of the form

$$\begin{aligned} H_n^{\text{MMHK}} &= \sum_{\mathbf{k},\alpha,\alpha',\sigma} g_{\alpha\alpha'}(\mathbf{k})c^{\dagger}_{\mathbf{k}\alpha\sigma}c_{\mathbf{k}\alpha'\sigma} - \mu\sum_{\mathbf{k},\alpha}\hat{n}_{\mathbf{k}\alpha\sigma} \\ &\quad + \sum_{\mathbf{k},\alpha} U\hat{n}_{\mathbf{k}\alpha\uparrow}\hat{n}_{\mathbf{k}\alpha\downarrow}, \end{aligned} \tag{5}$$

where n is the number of mixed momenta, **k** is summed over rBZ$_n$ and $g(\mathbf{k})$ is the dispersion matrix whose elements in a tight-binding model are defined by $g_{\alpha\alpha'} = \sum_{\delta\mathbf{r}} t^{\alpha,\alpha'}_{\delta\mathbf{r}} e^{i\mathbf{k}\cdot(\mathbf{r}_\alpha - \mathbf{r}_{\alpha'} + \delta\mathbf{r})}$, where **r** denotes the position of the unit cell, $t^{\alpha,\alpha'}_{\delta\mathbf{r}}$ represents the hopping between sites α and α' located in unit cells separated by $\boldsymbol{\delta}\mathbf{r}$ and σ is the spin. Although the kinetic and interaction terms no longer commute, the Hamiltonian still maintains the form $\sum_{\mathbf{k}\in\text{rBZ}_n} h_{\mathbf{k}}$, is governed by the HK fixed point[29,40] and still remains exactly solvable when the unit cell is not too big. Since the Hamiltonian in equation (5) depends explicitly on n, we also denote it as the n-MMHK model. Of course, the solvability is now based on finding the eigenstates of an n-site cluster. Figure 1b depicts the reduction of rBZ when n sites decorate each **k** point (Extended Data Fig. 1 shows the real-space counterparts). In the limit of $n \to N$, the **k** summation vanishes reducing the interaction term entirely to

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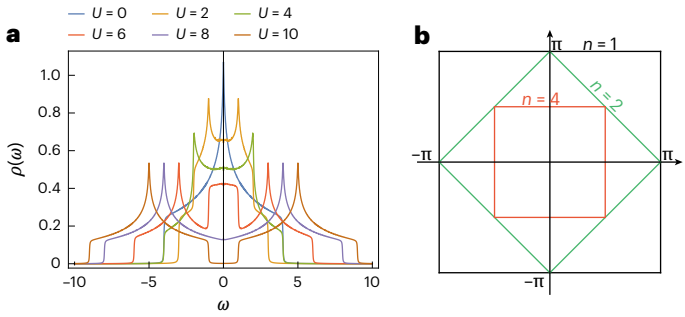


Fig. 1 | Mott transition in band HK model and scheme for the *n*-MMHK model. **a**, DOS under different U values displaying the Mott transition in the half-filled band HK model at inverse temperature $\beta = 200/t$. **b**, Evolution of the rBZ as the number of mixed momenta n increases, leading to a purely local (in real space) model when $n = N$.

$$\lim_{n\to N}\sum_{\alpha=1}^{n}\sum_{\mathbf{k}\in \mathrm{rBZ}_n} U n_{\mathbf{k}\alpha\uparrow} n_{\mathbf{k}\alpha\downarrow} = \sum_{\alpha=1}^{N} U n_{\alpha\uparrow} n_{\alpha\downarrow}, \tag{6}$$

which is just a summation over the local degrees of freedom as in the on-site Hubbard model. Hence, we are guaranteed to obtain Hubbard physics when n is sufficiently large.

One dimension

Figure 2a presents a comparison of the ground-state energy versus U between the exact Bethe ansatz result of the one-dimensional (1D) Hubbard model and that of n-MMHK with varying n. As n increases, the n-MMHK model rapidly approaches the exact result[1]. Already at $n = 10$, the deviation from the exact result is less than 1%. Note that even for relatively small n, it converges to the exact Hubbard results consistently as n increases across all U values. This behaviour stands in sharp contrast to finite Hubbard chain calculations, where deviations from the exact results vary unpredictably with both chain length L and U when L is small (Extended Data Fig. 2). This suggests that a small n can capture the essential Hubbard physics qualitatively, with increasing n offering a systematic path to quantitative accuracy.

To further quantify the rate of convergence with increasing n, we solve the n-MMHK model using DMRG with n as large as 40. In Fig. 2b, we plot the difference between the n-MMHK ground-state energy and the exact Bethe ansatz result for the Hubbard model in the TD limit as a function of $1/n$ for $U/t = 4, 6, 8$. Rather than scaling as $1/n$, we find that the actual convergence follows a power law $1/n^{\gamma}$ with $\gamma \approx 2$. Specifically, $\gamma = 1.83$ for $U = 4$, increasing with U and reaching $\gamma = 2.07$ for $U = 8$. By contrast, the standard finite-sized Hubbard chain calculation using DMRG with the open-boundary condition only shows a scaling of $1/L$ (Extended Data Fig. 3). These results indicate that the n-MMHK model offers a more rapid convergence towards Hubbard physics in the TD limit than finite-cluster Hubbard calculations. In particular, we later observe an even faster convergence rate than $1/n^2$ when extending the n-MMHK framework to a quasi-2D ladder geometry (Extended Data Fig. 4). This consistent convergence behaviour enables data-driven techniques to efficiently extrapolate to the $n \to N$ limit. We also compare the single-particle charge or Mott gap, Δ, between both n-MMHK-DMRG and Hubbard-DMRG (Extended Data Fig. 5). In n-MMHK, Δ vanishes when $U \to 0$, as expected, whereas it remains finite in Hubbard-DMRG as a result of finite-size effects. This limitation arises because the open-boundary condition, regardless of the system size, cannot sample the $\mathbf{k} = 0$ point, which is the momentum at which the gap is minimized in $d = 1$. By contrast, n-MMHK is constructed to directly represent the TD limit, and as a result, it correctly captures the vanishing of the gap as $U \to 0$. Finally, the n-MMHK-DMRG calculations involve a smaller number of mixed momenta compared with the standard Hubbard-DMRG chain for the same accuracy, due to the intrinsic periodic-boundary conditions imposed by the n-MMHK Hamiltonian. Since DMRG performs worse with periodic-boundary conditions (typically requiring about five times the bond dimension compared with the open-boundary condition), this requirement limits the accessible number of mixed momenta. Nonetheless, our 1D study provides a clear and controlled demonstration of the n-MMHK model's rapid convergence. In the following sections, we turn to the real challenge: applying n-MMHK to two-dimensional (2D) systems.

Two dimensions

Applying the n-MMHK model to 2D Hubbard simply requires arranging the n sites into a cluster. The consistent and rapid convergence observed in 1D chain and quasi-2D laddder give us reason to expect that even a small number of sites can capture essential Hubbard physics, independent of dimensionality. In the following, we confirm this idea by applying the 4-MMHK model (where '4' denotes the number of mixed momenta) to two dimensions, and demonstrate that further quantitative accuracy in the spectral function is achieved with the 16-MMHK model (with a 4×4 cluster for each $\mathbf{k}$). For $n = 4$, implemented here as a 2×2 cluster, the matrix $g(\mathbf{k})$ in equation (5) is

$$\begin{pmatrix} 0 & \varepsilon_{tx} & \varepsilon_{ty} & \varepsilon_{t'} \\ \varepsilon_{tx} & 0 & \varepsilon_{t'} & \varepsilon_{ty} \\ \varepsilon_{ty} & \varepsilon_{t'} & 0 & \varepsilon_{tx} \\ \varepsilon_{t'} & \varepsilon_{ty} & \varepsilon_{tx} & 0 \end{pmatrix}, \tag{7}$$

where $\varepsilon_{tx(y)} = -2t\cos k_{x(y)}$; $\varepsilon_{t'} = -4t'\cos k_x \cos k_y$; and t and t' represent the first- and second-neighbour hopping, respectively. The matrix $g(\mathbf{k})$ for a 3×3 cluster (9-MMHK) is given in Supplementary Equation (1) and Supplementary Fig. 1, and is straightforward to generalize to any cluster. We set $t' = 0$, unless otherwise specified. This matrix is identical in form to that of a 2×2 Hubbard cluster with twisted-boundary conditions[47], where α is replaced with the site index i and $k_x = \theta_x$ and $k_y = \theta_y$, where θ_x and θ_y define the twisted-boundary condition. With n-MMHK, however, the twist has a physical interpretation as the crystal momentum, thereby placing n-MMHK in the TD limit.

We compute the DOS of half-filled 4-MMHK for a square lattice at $t' = 0$ and find that any non-zero U is sufficient to eliminate the metallic state, producing a vanishing DOS at zero energy (Fig. 3a). This reflects the instability caused by Fermi surface nesting. This is captured already in 4-MMHK—a qualitative improvement from band HK in which $U_c = W$ (bandwidth)—and is missed by DMFT[2], though cluster DMFT[11] progressively yields a smaller U_c as the cluster size increases, consistent with

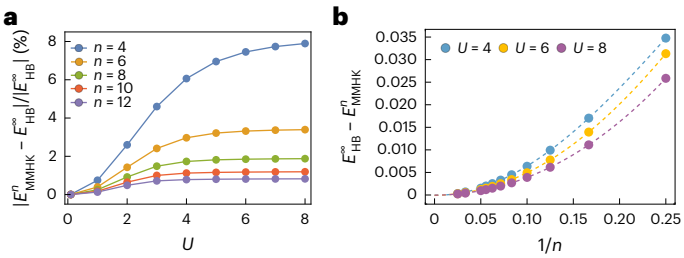


Fig. 2 | Ground-state energy deviation of *n*-MMHK from Bethe ansatz. Comparison of the difference between the ground-state energy with n-MMHK model, E^n_{HK}, and the infinite-system size Hubbard Bethe ansatz energy, E^{∞}_{HB}, both in one dimension. **a**, n-MMHK is solved by ED. **b**, n-MMHK is solved by DMRG including as many as $n = 40$ mixed momenta at various U values. The dashed lines are polynomial regression fitting with extrapolation to $1/n \to 0$ ($n \to \infty$). The fitting curves can be well represented as $f(U=4) = 0.45(1/n^{1.83}) + a$, $f(U=6) = 0.51(1/n^{2.01}) + a$ and $f(U=8) = 0.45(1/n^{2.07}) + a$. The asymtotic values a at $1/n = 0$ are -5.4×10^{-5} ($U = 4$), -9×10^{-5} ($U = 6$) and -4×10^{-5} ($U = 8$).

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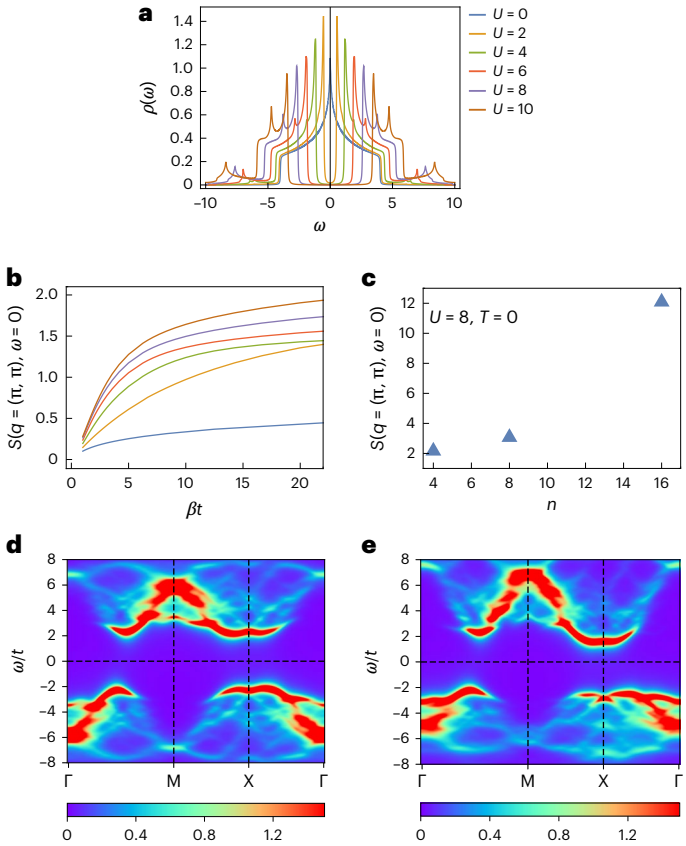


Fig. 3 | Mott transition, AF susceptibility and spectral functions for the half-filled *n*-MMHK model. a, DOS representing the Mott transition for the half-filled 4-MMHK model at $\beta = 200/t$. **b**, AF spin susceptibility of the half-filled 4-MMHK model plotted as a function β for various U values (same legend as in **a**). **c**, Zero-temperature limit at $U = 8$ for 4-, 8- and 16-MMHK. **d**,**e**, Spectral function $A(k, \omega)$ at half-filling of the 16-MMHK model at $t' = 0$ (**d**) and $t'/t = -0.25$ (**e**) with $U = 8$, zero temperature and a broadening factor of 0.2.

the results from dynamical cluster approximation[13] and dynamical vertex approximation[18]. For finite t', which breaks nesting, a finite U_c is needed to open a charge gap (Extended Data Fig. 6). Another substantial improvement over the band HK model, which exhibits a diverging ferromagnetic susceptibility, is that the 4-MMHK model shows antiferromagnetic (AF) correlations as the leading spin response (Extended Data Fig. 7), a key feature of Hubbard physics. In Fig. 3b, we show that the AF spin susceptibility $S(q = (\pi, \pi), \omega = 0) = \int_0^\beta d\tau \langle \hat{S}^z(q, \tau)\hat{S}^z(0)\rangle$ increases with increasing U and decreasing temperature. The Mott insulating state can be reached either by lowering the temperature at large U or by increasing U at low temperature. In two dimensions, AF ordering, prohibited at finite temperature by the Mermin–Wagner theorem[48], is not crucial to the temperature-driven transition at energy scales of U, well above the AF exchange scale of t^2/U. By contrast, at low temperature with finite t', the AF susceptibility shows a cusp at U_c (Extended Data Fig. 6), indicating that AF fluctuations are present in the interaction-driven Mott transition. The persistence of AF correlations away from half-filling (Extended Data Fig. 7) further highlights their key role in doped Hubbard physics. The AF spin susceptibility (Fig. 3b) is underestimated in the 4-MMHK model—where the system consists of Hubbard squares in **k** space—compared with QMC results on larger Hubbard clusters[19,49], and it does not diverge at zero temperature. This quantitative discrepancy can be mitigated by including additional mixed momenta (Fig. 3c). We then compute the spectral function $A(k, \omega)$ at $U/t = 8$ of the half-filled 16-MMHK (with a 4×4 cluster per **k**) using Lanczos exact diagonalization (ED) for $t'/t = 0$ and -0.25 (Fig. 3d,e, respectively). The narrowing dispersion of the lower and upper Hubbard bands are in quantitative agreement with state-of-the-art QMC and cluster perturbation theory for the 2D Hubbard model[34,35,50] at $t' = 0$ (Extended Data Fig. 8 shows the 4- and 8-MMHK results, which already capture the qualitative features). For $t' = -0.25$, unbiased QMC suffers from the sign problem at low temperatures. Our result (Fig. 3e) exhibits particle–hole asymmetry and an indirect Mott gap. This indicates that doped holes preferably occupy regions around $(\pi/2, \pi/2)$, whereas doped electrons appear first near $(0, \pi)$ and $(\pi, 0)$, laying a solid foundation for future studies on particle–hole asymmetric pseudogap physics.

Given that the kinetic and potential terms do not commute, the number of doubly occupied sites plays a crucial role in all the ground-state properties. To this end, we compute the double occupancy for the *n*-MMHK defined as

$$D_n = \frac{1}{N} \sum_{\mathbf{k} \in \mathrm{rBZ}_n, \alpha} \langle n_{\mathbf{k}\alpha\uparrow} n_{\mathbf{k}\alpha\downarrow} \rangle, \tag{8}$$

which is directly related to the interaction energy. Consider the stark difference between the non-interacting limits of the band HK and *n*-MMHK models in terms of double occupancy. Regardless of the model, in the non-interacting limit, the expression for D_n factorizes as

$$D_n^{\text{non-int}} = \frac{1}{N} \sum_{\mathbf{k} \in \mathrm{rBZ}_n, \alpha} \langle n_{\mathbf{k}\alpha\uparrow} \rangle \langle n_{\mathbf{k}\alpha\downarrow} \rangle. \tag{9}$$

For the band HK model ($n = 1$), half the BZ is doubly occupied. That is, $\langle n_{\mathbf{k}\alpha\sigma} \rangle = 1$ only if $\mathbf{k}\alpha$ is occupied. Otherwise, $\langle n_{\mathbf{k}\alpha\sigma} \rangle = 0$. Consequently, $D_1^{\text{non-int}} = 1/2$. On the other hand, for the *n*-MMHK model, $\langle n_{\mathbf{k}\alpha\sigma} \rangle = \langle n_{\mathbf{k}\beta\sigma} \rangle$ for all $\alpha \neq \beta$ by rotational symmetry. In addition, $\langle n_{\mathbf{k}\alpha\sigma} \rangle = 1/2$ in the full range of the rBZ. As a result, we find that for the *n*-MMHK model, $D_{n>1}^{\text{non-int}} = 1/4$. Note that this result is identical to the non-interacting limit of the Hubbard model and underscores how drastically band HK differs from its momentum-mixing counterpart. Figure 4a for D_n shows this dramatic difference. In band HK, D_1 decreases steadily from 1/2 and vanishes for $U > W$. However, in the *n*-MMHK model, D_n, starting at 1/4, just tapers asymptotically as U increases. Note also the rapid convergence between the $n = 2$ and $n = 4$ cases, which matches closely with state-of-the-art auxiliary-field QMC simulations (AFQMC) (H. Xiu et al., unpublished data) and finite-temperature QMC (FTQMC)[33] on the Hubbard model. The inset also shows the double occupancy of the 4-MMHK model at 1/8 hole doping ($x = 0.125$) and its benchmark with QMC simulations (H. Xiu et al., unpublished data) (Supplementary Information provides an explanation for the slight mismatch at $U = 0$ due to definition). This agreement here underscores that *n*-MMHK converges rapidly to Hubbard physics with negligible computation effort.

The fact that the double occupancy only vanishes asymptotically as U increases in the *n*-MMHK model reflects the dynamical mixing between the upper and lower Mott sub-bands. This arises entirely from the non-commutativity[36,37,51] of the kinetic and potential energies and appears as t/U corrections to the low-energy spectral weight (LESW). In the atomic limit, the LESW is strictly $2x$ (where x is the doping level) because each hole can be occupied by a spin-up or spin-down electron. Any dynamical corrections to this necessarily generate double occupancy and, hence, increase the LESW strictly defined as

$$\Lambda(x) \equiv \mathrm{LESW} = \int_0^{\omega_g} N(\omega) \mathrm{d}\omega. \tag{10}$$

The ω_g is located at the spectral gap in the DOS (Extended Data Fig. 9). In Fig. 4b, we compare the LESW of the 4-MMHK model with figure 3 of ref. 36. There is no qualitative difference with ED on the (1D) Hubbard model[36,37]. Both increase faster than $2x$. The semiconductor (dashed

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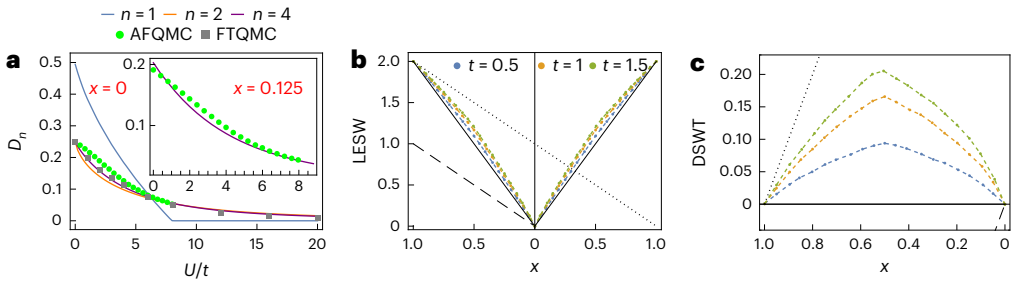


Fig. 4 | Double occupancy and DSWT for *n*-MMHK. a, D_n at half-filling ($x = 0$) as a function of U for the n-MMHK model with various n values. The green and grey dots denote the AFQMC (H. Xiu et al., unpublished data) and FTQMC[33] calculations of the standard double occupancy of the 2D Hubbard model. The inset in **a** compares the 4-MMHK and Hubbard AFQMC results (H. Xiu et al., unpublished data) at 1/8 hole doping ($x = 0.125$). **b**,**c**, LESW (**b**) and DSWT (**c**) are the exact solutions of the 4-MMHK model ($U = 10$) for the hopping parameters at $\beta = 30$. The solid line shown with a $2x$ slope is the band HK or atomic Hubbard result. The dashed and dotted lines depict the semiconductor and Fermi liquid results, respectively. Note that there is no qualitative difference with the ED results for the Hubbard model[36,37]. As t/U increases, so does the DSWT.

line) and Fermi liquid (dotted line) results are shown for comparison. As the occupied part of the lower band has a weight $1 - x$, the total weight in the lower band now exceeds $1 + x$. As only $1 + x$ electrons can occupy the lower band, DSWT, defined as $\Lambda(x) - 2x$ and plotted in Fig. 4c, implies[52] that the spectral weight in the lower band cannot be exhausted by counting electrons alone. Figure 4c is quantitatively in agreement with the Hubbard model (figure 4 of ref. 36), the only difference being the maximum value for which 4-MMHK is 0.2, whereas it is 0.23 for the (1D) Hubbard result. Consequently, the fact that the LESW in the n-MMHK model increases faster than $2x$ is a profoundly non-trivial result as no analytically solvable model has ever captured this feature of the Hubbard model. The ability of the n-MMHK model to reproduce this behaviour with as few as $n = 2$ or $n = 4$ mixed momenta offers a rare opportunity to gain an analytical insight into DSWT, as demonstrated in a follow-up study[53].

Aside from DSWT, n-MMHK also exhibits a pseudogap in the DOS. Figure 5 shows the DOS for the 4-MMHK model with $t' = -0.25$ (Fig. 5a,c) and $t' = 0$ (Fig. 5b,d). Figure 5a shows a suppression of the DOS at the chemical potential ($\rho(\omega = 0) \lesssim 0.1$) in the underdoped region with $t' = -0.25$ when $U \geq W$, indicative of a pseudogap in the absence of a superconducting order. The complete tracking of the DOS at zero frequency (Fig. 5c) displays the trend that as U increases, the pseudogap region appears and extends to a higher hole-doped density. By contrast, when $t' = 0$, the pseudogap suppression does not arise until $U \gg W$ (Fig. 5b,d). These observations are consistent with a recent study[47] on a 2×2 Hubbard cluster with the twisted-boundary condition. The emergence of the pseudogap on including these scattering momenta highlights their crucial role in suppressing spectral weight, consistent with a recent solvable model construction[54].

Finally, we compute the heat capacity of the 4-MMHK model at half-filling (Fig. 6). Additional results for the 1- and 2-MMHK cases are provided in Extended Data Fig. 10. Most noticeable is the two-peak structure at $U > W$, representing a demarcation of the charge and spin degrees of freedom into the high- and low-temperature regimes, respectively, consistent with the QMC simulations on the Hubbard model[38,39]. At half-filling, we find a near-crossing of the heat capacity curves as a function of U at a temperature intermediate between the spin and charge excitations. The Maxwell relations governing the entropy dictate[55] the existence of such a crossing. Since there is no sign-problem restriction, we can access the low-temperature heat capacity exactly. As shown in Fig. 6b, the low-temperature heat capacity data follow a power-law increase (Supplementary Information). Although such algebraic growth might seem counterintuitive for a Mott insulator, the first-excited state is charge neutral[56,57] (Supplementary Information). Such charge-neutral excitations determine the low-temperature behaviour of the specific heat and are consistent with the absence of long-range magnetic order.

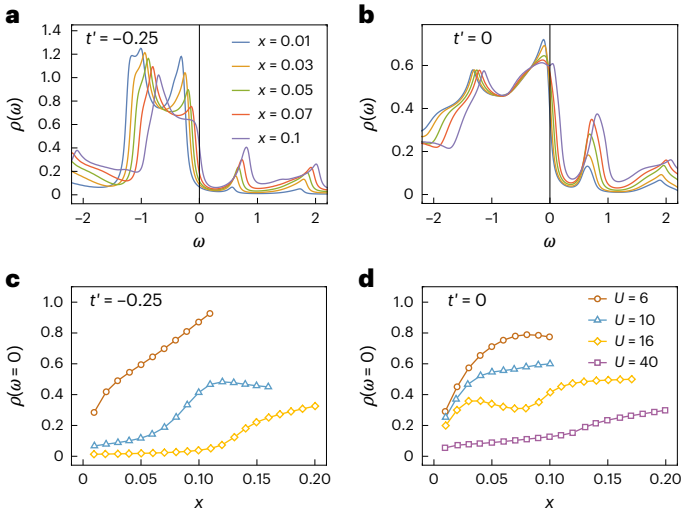


Fig. 5 | Observation of a pseudogap in 4-MMHK. a,**b**, DOS at varying hole-doped densities with (**a**) and without (**b**) the next-nearest-neighbour hopping t' for the 4-MMHK model ($U/t = 10$, $\beta = 30/t$). **a** and **b** share the same legend. Only for $t' \neq 0$, $\rho(\omega)$ is suppressed at zero frequency in the underdoped region, thereby indicating a pseudogap. **c**,**d**, $\rho(\omega = 0)$ with $t' = -0.25$ (**c**) and $t' = 0$ (**d**) as a function of hole-doped density under various U values.

Scaling

The hidden power of the n-MMHK model lies in its rapid and consistent convergence towards Hubbard physics. This efficiency arises from the number of degrees of freedom introduced by the site number at each **k** point: n sites generate n hybridized momenta, leading to n^2 interaction terms at each **k** point (equation (3)) and, hence, the scaling $1/n^2$, independent of dimensionality. Including momentum mixing in this way effectively introduces quantum fluctuations. We demonstrate the $1/n^2$ scaling in one dimension (Fig. 2), where quantum fluctuations are the strongest. In Extended Data Fig. 4, we find even faster convergence ($\sim 1/n^3$) on a quasi-2D ladder system, particularly at larger U values, as a result of suppressed fluctuations in higher dimensions. This makes two dimensions an ideal setting for the MMHK framework: a computable number of sites can cover a square or rectangular unit cell with a sizable linear extent l, whereas quantum fluctuations remain moderate. Furthermore, this argument points towards a monotonically increasing exponent $\gamma(d)$ as a function of the lattice dimension giving rise to a convergence of $1/n^{\gamma(d)}$, where $\gamma(d = 1) = 2$ and increases for $d > 1$. Therefore, vanishingly few mixed momenta are required in $d = \infty$ to achieve convergence, implying a consilience between the band HK model and the Hubbard model in this limit. As the underlying Mott transition is

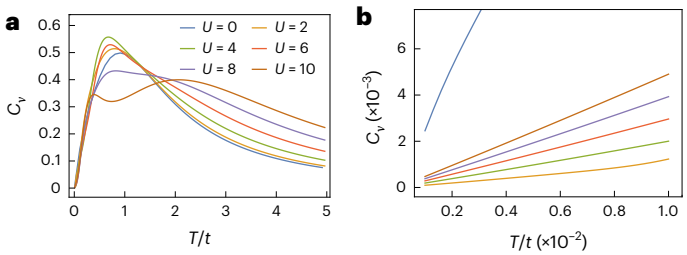


Fig. 6 | Heat capacity of the half-filled 4-MMHK model. a,**b**, Heat capacity as a function of temperature at high (**a**) and low (**b**) temperatures for various U values. **a** and **b** share the same legend.

governed by the breaking of the Z_2 symmetry[29], thereby determining the universality class of the second-order end point, this rapid convergence in $d = \infty$ is highly suggestive of the similarity in criticality between HK and DMFT[2]. Since this involves an in-depth comparison of the critical exponents, we postpone this for a future study.

Why does this scheme work so well? We have previously shown[29] using the Bott topological invariant that the surface of zeros or the Luttinger surface—the defining feature that leads to the charge gap in a Mott insulator—is stable against any continuous local deformation. Since the momentum-mixing construction is based on a n-site extension, thereby a local deformation, the Luttinger surface must exist for any n. Consequently, as far as the charge dynamics are concerned, HK and Hubbard lie in the same universality class. This perspective is valid because they are both governed by the Mott fixed point of $\mathbb{Z}_2$ symmetry breaking[29]. The spectral weight bifurcation into the upper and lower Hubbard bands is preserved under momentum-mixing extension[40], allowing n-MMHK to faithfully capture the essential charge physics of the Hubbard model and incorporating non-trivial dynamics in a controlled and systematic way.

Concluding remarks

Ultimately, the deep connection between n-MMHK and Hubbard physics stems from the resilience of HK to local continuous deformations. The momentum-mixing construction represents such a local deformation. Since these deformations leave[29] the Luttinger surface intact, the same universality class governs the charge dynamics that mitigate the Mott transition in both n-MMHK and Hubbard models. This applies only to the charge sector; differences arise in the spin sector. For example, although band HK exhibits a ferromagnetic instability[42], the n-MMHK and Hubbard models both display antiferromagnetism. This is expected as the $\mathbb{Z}_2$-breaking fixed point only controls the charge dynamics. After all, it is the charge gap that is the universal feature of all Mott insulators. This is consistent with Mott's original idea[58] that the spin sector is ancillary to the insulating ground state in partially filled bands. In fact, our fixed-point argument[40] is a proof of Mott's idea and n-MMHK is the numerical demonstration of its validity.

We have shown that the n-MMHK model captures the key **k**-summed quantities such as the DOS, reflecting the underlying Mott fixed point. This suggests that even with a small number of mixed momenta, n-MMHK can potentially provide a meaningful insight into broader phenomena like the strange-metal phase. For **k**-resolved properties, $n = 16$ yields quantitatively accurate spectral functions, enabling detailed studies of Fermi surface reconstruction and the pseudogap. At this scale, spin fluctuations relevant to superconductivity and stripe order are also captured quantitatively. Additionally, n-MMHK can be used to study transport provided the order of limits is correctly[32] taken (1) $q \to 0$ and then (2) $L \to \infty$ rather than the inverse, which yields erroneous[59] results. In summary, the rapid and systematic convergence of n-MMHK to Hubbard physics makes it a powerful and efficient framework for exploring strong correlations. With just a few mixed momenta, it offers a simplified yet faithful platform to study the interplay between correlations and other effects such as topology and non-equilibrium dynamics.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41567-025-03095-1.

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Methods

DMRG

DMRG[60,61] is used to solve n-MMHK in one dimension and a quasi-2D ladder. Equation (5) can be expressed as a sum over $\mathbf{k}$ for the n-site Hubbard model with a complex hopping matrix $g_{\alpha\alpha'}(\mathbf{k})$. DMRG is then performed for each of these Hubbard systems (for each $\mathbf{k}$ value) and the MMHK ground-state energy is computed by averging over the ground-state energies of each Hubbard cluster.

ED

ED with the Lanczos algorithm[62] is used to determine the ground state of the 16-MMHK model at zero temperature and to compute the dynamical quantities including the spectral function and spin susceptibility. For this model, we need to solve the 16-site Hubbard cluster for each $\mathbf{k}$ (400 clusters in total). For each 16-site Hubbard cluster, we use a Lanczos iteration to solve for the ground state at each total momentum (16 in total) with an energy accuracy of ~10^{-12} and then locate the true ground state among them. On the basis of that state, we then conduct another Lanczos iteration to estimate the dynamical quantities.

Data availability

The data and scripts needed to reproduce the figures are openly available via Zenodo at https://doi.org/10.5281/zenodo.17096693 (ref. 64). Source data are provided with this paper.

Code availability

The ED code used for solving the 16-MMHK model is available via GitHub at https://github.com/wztzjhn/quantum_basis. The DMRG algorithm is implemented through TeNPy, which can be accessed at https://tenpy.readthedocs.io/en/v1.0.6/reference/tenpy.algorithms.dmrg.html. All other codes are available from the corresponding authors upon reasonable request.

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Acknowledgements

We thank B. Bradlyn for helpful comments, S. Zhang for sending us his double-occupancy data (Fig. 4a, green dots), G. Sawatzky for a helpful e-mail exchange and Z. Wang for his help in ED. This work was supported by the Center for Quantum Sensing and Quantum Materials, a Department of Energy, Energy Frontier Research Center, grant number DE-SC0021238 (P.M. and P.W.P.). P.W.P. also acknowledges NSF DMR-2111379 for partial funding of the HK work, which led to these results. P.M. was also supported by the Gordon and Betty Moore Foundation's EPiQS Initiative through grant number GBMF 8691. The DQMC calculation of this work used the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) Expanse supercomputer through the research allocation TG-PHY220042, which is supported by National Science Foundation via grant number ACI-1548562 (ref. 63).

Author contributions

P.M. performed the calculations for the 4- and 16-MMHK models in two dimensions and analysed the data. J.Z. performed the calculations for the 2-MMHK model and carried out the analytic derivation. G.T. performed the DMRG calculations for MMHK in one dimension and ladder. N.A.H. performed the ED calculations for MMHK in one dimension. D.K. performed the calculations for local correlations in the band HK model. D.P. performed the calculations for spin correlations in the band HK model. P.W.P. supervised the project. P.M., J.Z., G.T. and P.W.P. wrote the paper with input from all authors.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41567-025-03095-1.

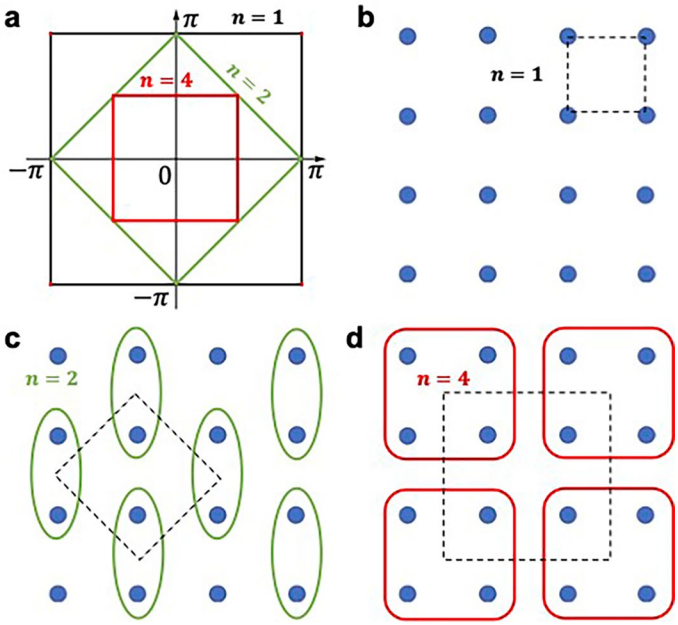
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41567-025-03095-1.

Correspondence and requests for materials should be addressed to Peizhi Mai or Philip W. Phillips.

Peer review information *Nature Physics* thanks the anonymous reviewers for their contribution to the peer review of this work.

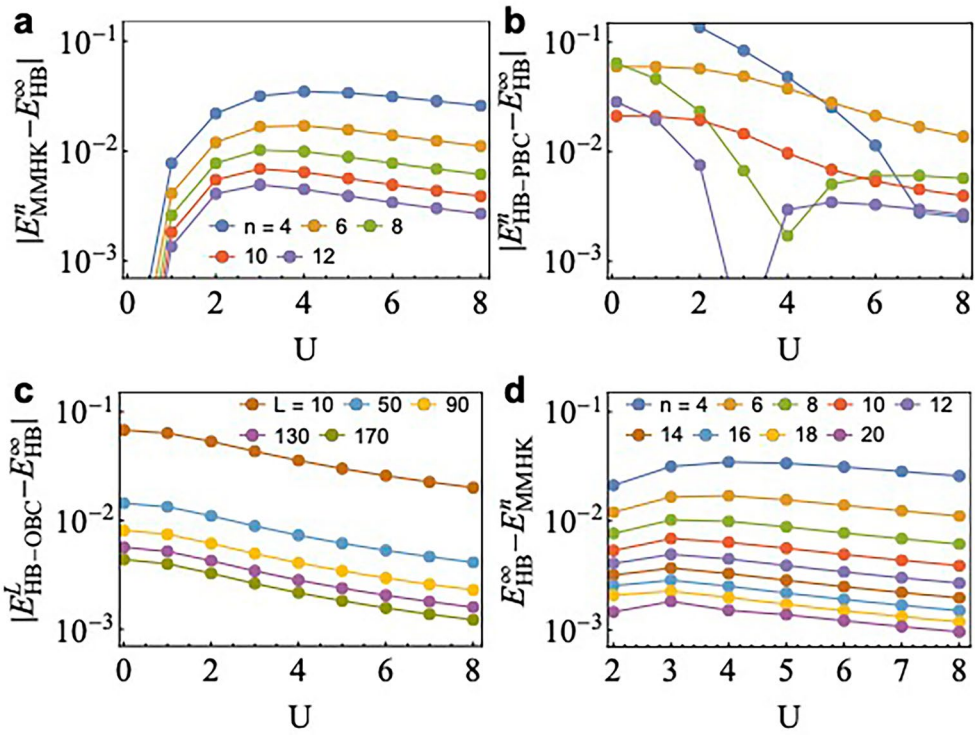
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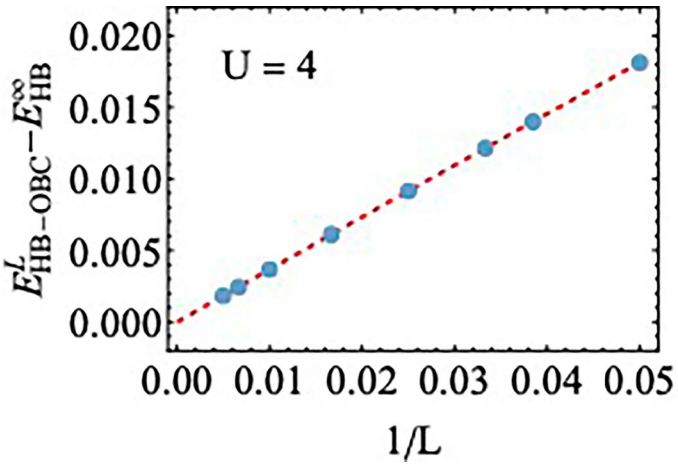
Extended Data Fig. 1 | Scheme for implementing *n*-MMHK in two dimensions. **a** is reproduced from Fig. 1b for convenience. Alternative real-space construction by grouping the atoms into cells (encircled by green/red lines) with an updated lattice constant. **b**, Original unit cell in the band HK model with primitive unit cell in dashed line. **c**, The 2-MMHK model leads to a doubling of the unit cell size and the lattice constant (dashed line) **d**, The 4-MMHK model results in a quadrupling of the unit cell size. When $n = N$, the unit cell contains all the sites.

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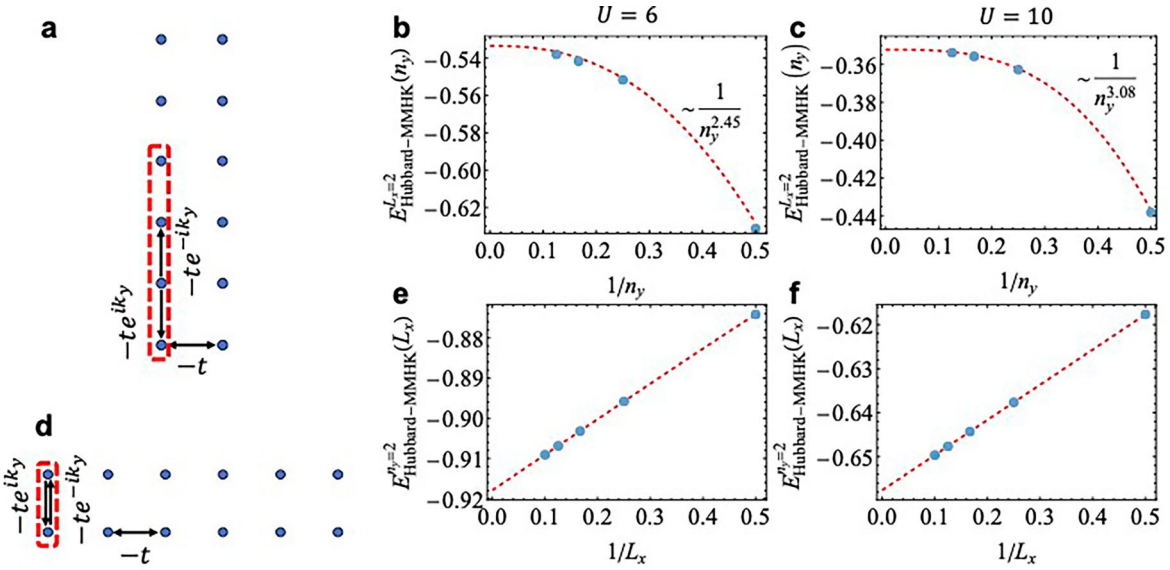
Extended Data Fig. 2 | Ground-state energy accuracy of *n*-MMHK and finite-size Hubbard clusters in one dimension. Comparison of the deviation in ground-state energy relative to the Bethe ansatz result for the TD limit (E^{∞}_{HB}) among different model settings. **a** shows the deviation of the *n*-MMHK model (E^{n}_{MMHK}), which is consistently less than 1% for $n \geq 10$. **b** gives the deviation of a periodic *n*-site Hubbard chain ($E^{n}_{\mathrm{HB-PBC}}$), solved by ED, with the same legend as **a**. The scattered *U*-dependence for various *n* indicates non-systematic finite-size effects. **c** displays the deviation of an open Hubbard chain with length *L* ($E^{L}_{\mathrm{HB-OBC}}$), obtained from standard DMRG. **d** is similar to **a** but with larger *n* (the number of mixed momenta) and the *n*-MMHK models are solved by DMRG for $n > 12$. Comparing **c** and **d**, we find that *n*-MMHK exhibits more stable accuracy across the range of *U*, while open Hubbard chains show larger errors for small *U*.

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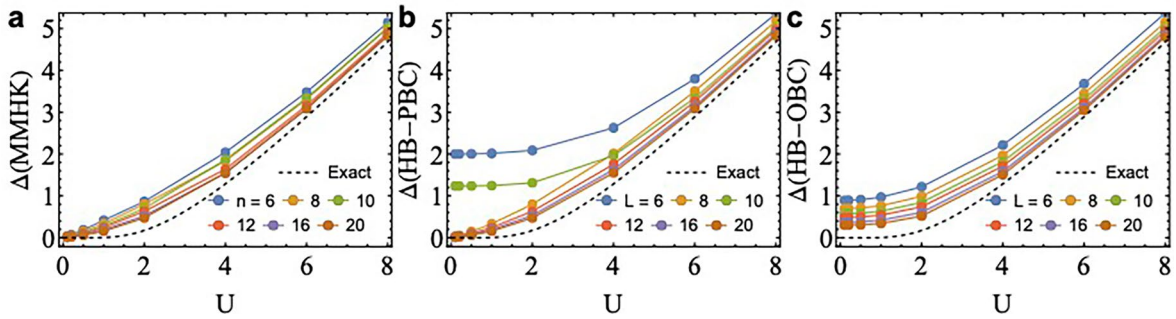
Extended Data Fig. 3 | Scaling of ground-state energy deviation of an open Hubbard chain from Bethe ansatz. Standard DMRG results for an open finite Hubbard chain at $U = 4$ are shown as deviations from the Bethe-ansatz ground-state energy. The deviation scales linearly with $1/L$.

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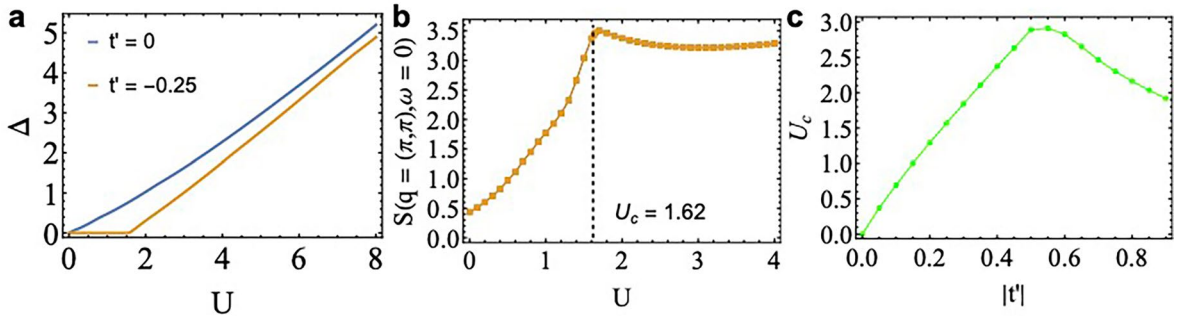
Extended Data Fig. 4 | Two-leg Hubbard-MMHK hybrid ladder. The hybrid ladder consists of sites along along the y-direction which was solved using MMHK and sites along the x-direction described by Hubbard. We used open boundary conditions along the x-direction. The physical extent along (**a**) the *y*-direction and (**d**) the *x*-direction is clearly shown. The ground-state energy of the hybrid ladder is shown for $U = 6$ (smaller than bandwidth $w = 8$) in (**b**, **e**) and for $U = 10$ (larger than w) in (**c**, **f**). **b**, **c** display the ground-state energy as a function of n_y, the number of MMHK sites along the *y*-direction, corresponding to the setup in (**a**). **e**, **f** show the energy as a function of L_x, the length along the *x*-direction, corresponding to the setup in (**d**). These results are all obtained using DMRG and the dashed lines represent a fitting curve. **b**, **c**, The energy converges faster than the $1/n^2$ behavior observed in the purely one-dimensional *n*-MMHK cases. **e**, **f**, The energy converges linearly as L_x increases.

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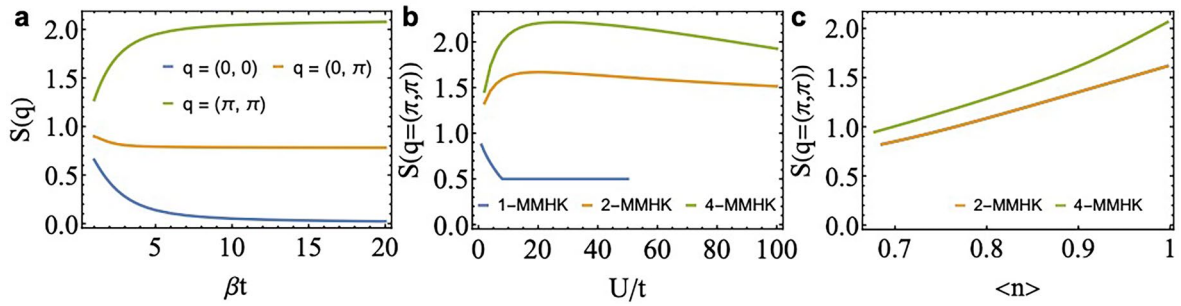
Extended Data Fig. 5 | Single-particle charge gap for *n*-MMHK, periodic, and open Hubbard chains in one dimension. The single-particle charge gaps are shown for (**a**) the *n*-MMHK model, (**b**) a periodic Hubbard chain with length *L*, and (**c**) an open Hubbard chain with length *L*, all obtained from DMRG simulations. In all panels, the dashed line denotes the exact result from Bethe ansatz. **a**, *n*-MMHK captures the vanishing gap at $U = 0$ (even for small n) and slowly converges to the exponential singularity (Bethe ansatz) at small U as n increases. **b**, Periodic Hubbard chain exhibits non-systematic L-dependence, with large L behaviors close to MMHK. **c**, Open Hubbard chain captures the larger-U regime accurately but slowly converges to the correct $U = 0$ limit due to finite length.

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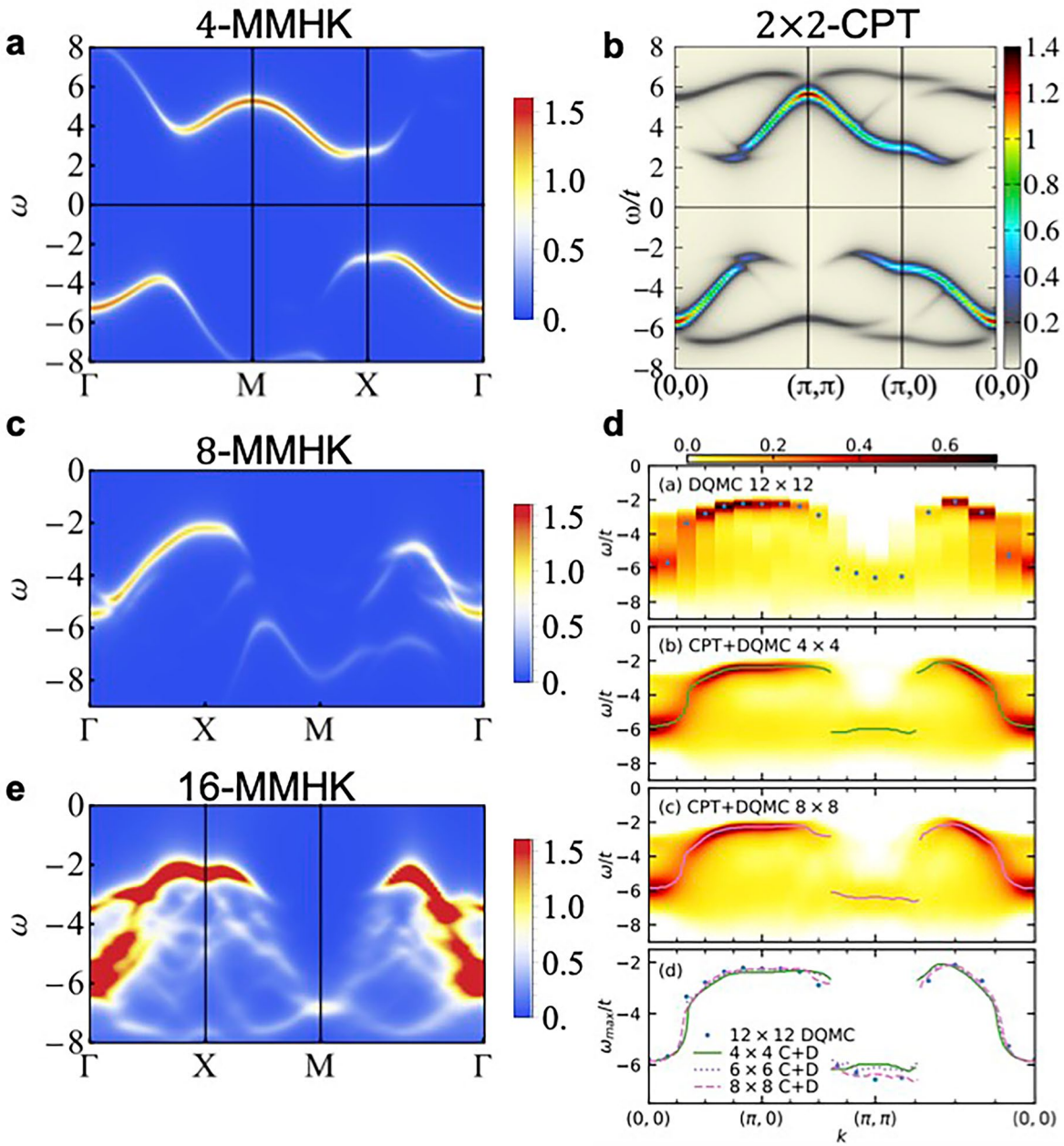
Extended Data Fig. 6 | Mott transition in the 4-MMHK model. a, Evolution of the csingle-particle charge gap Δ with U at half-filling for $t' = 0$ and $t'/t = -0.25$. At $t' = 0$, the gap opens for any finite U, while in the frustrated case ($t'/t = -0.25$), the gap opens only for $U > U_c = 1.62$, indicating a Mott transition. **b**, Antiferromagnetic spin susceptibility at $t' = -0.25$ and $\beta = 100$ as a function of U, showing a cusp at U_c. **c** The critical interaction strength U_c for the Mott transition as a function of $|t'|$ (the sign of t' does not make a difference in U_c). Initially, U_c grows almost linearly with $|t'|$. It saturates around $|t'| \approx 0.55$ and then decreases with further increasing $|t'|$.

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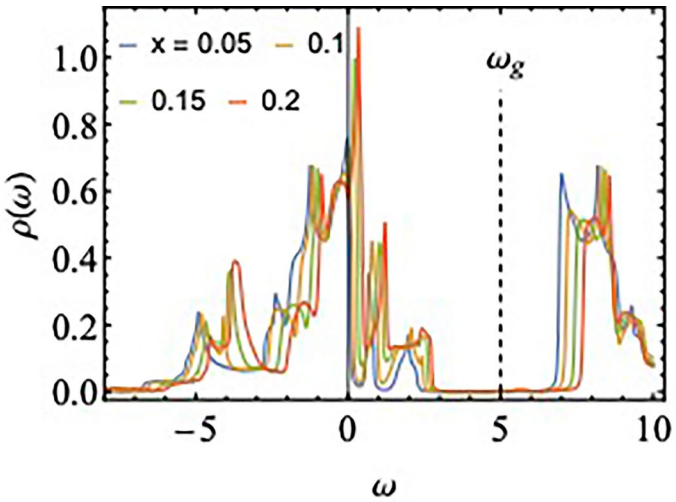
Extended Data Fig. 7 | Equal-time spin correlation for *n*-MMHK model. a, Equal-time spin correlation as a function of inverse temperature for different transfer momenta q for 4-MMHK at $U = 10$. The $q = (\pi, \pi)$ case dominates the spin correlation, indicating the leading magnetic fluctuation is of the antiferromagnetic type. **b**, Equal-time antiferromagnetic correlation as a function of U for different n-MMHK models. As the number of mixed momenta n increases from 1 (band HK), the antiferromagnetic correlation dominates. **c**, Equal-time antiferromagnetic correlation as a function of density at $U = 10$, $\beta t = 20$ for 2- and 4-MMHK models. The antiferromagnetic correlation remains for a wide range of doping, essential for capturing the more complicated doping physics in the Hubbard model.

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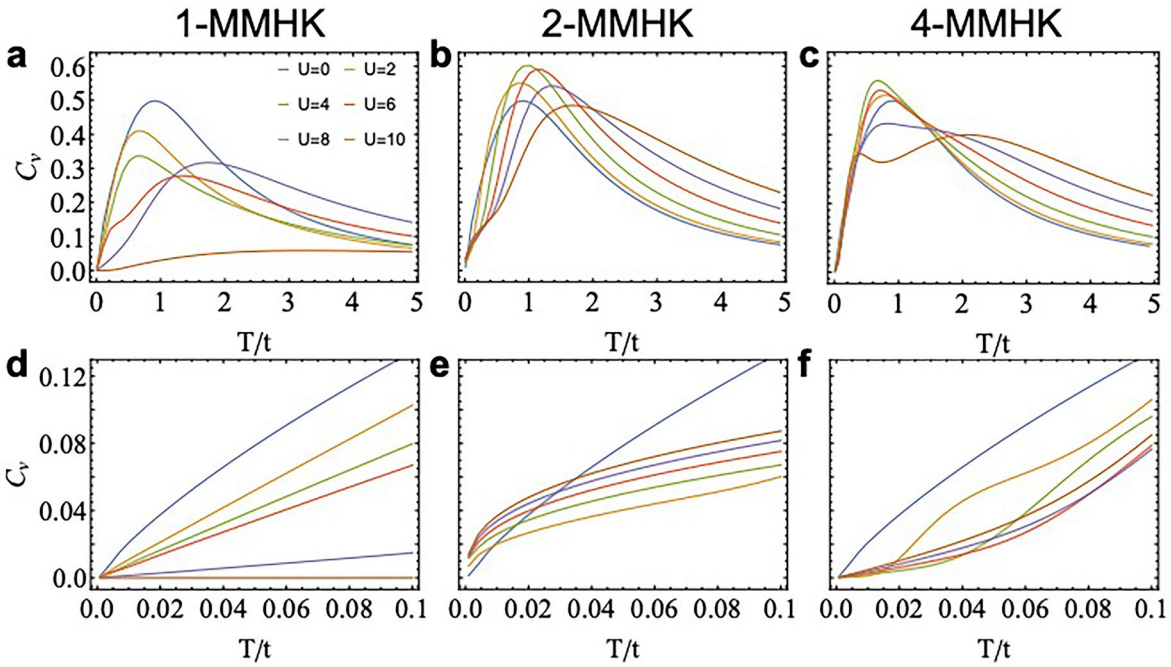
Extended Data Fig. 8 | Spectral functions of *n*-MMHK models with various *n*. We compare the half-filled ($U = 8$) spectral function $A(\mathbf{k}, \omega)$ among n-MMHK models (at zero temperature with a Lorentzian broadening of 0.2) and other methods for solving the Hubbard model. (**a**) 4-MMHK (or 2 × 2-MMHK model) and (**b**) 2 × 2 cluster perturbation theory (reprinted from[34]) show qualitatively similar leading features, including peak positions and gap size. The (**c**) 8-MMHK and (**e**) 16-MMHK results progressively improve the description of weaker sub-leading features, becoming quantitatively comparable to (**d**) state-of-the-art determinant quantum Monte-carlo and/or cluster perturbation theory (reprinted from[35]) at $\beta = 16$. This underscores that as $n \to N$, MMHK becomes the Hubbard model.

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Extended Data Fig. 9 | Density of states of the hole-doped 4-MMHK model and the choice of ω_g. We present the density of states at varying hole-doped densities x with $U = 10, \beta = 30$. The cutoff ω_g is set to the gap for calculating the low energy spectral weight with Eq. (10).

https://doi.org/10.1038/s41567-025-03095-1



Extended Data Fig. 10 | Heat capacity for *n*-MMHK models with *n* = 1, 2, 4 at various *U*. **a**-**c** are for high temperatures, while **d**-**f** are for low temperatures.